

Final Exam Part 1: Luxury Dealership Business Use Case

Project Overview + Context

Elite Motor Dealership is a luxury brand dealership located in Mishawaka, Indiana. Elite Motor Dealership is a full-service dealership offering their customers not only a full line-up of all new Lexus vehicles, but also numerous luxury and mid-range vehicles from similar brands. With a friendly and reliable service and parts department as well, Elite Motor Dealership has led sales in the market for the past 25 years.

In order to ensure Elite Motor Dealership is accurately serving their market and the needs of their customers; Elite Motor Dealership has tasked their in-house Marketing and Information Specialist, YOU, with analyzing their historical data for insights that can't help them understand the trends in their past performance that they can maximize today.

Criteria for Success

The in-house Marketing and Information Specialist, YOU, will analyze the historical sales data provided by Elite Motor Dealership.

You will present:

1. Your queries using SQLite with Python from prompts provided in a VS Code Notebook.
2. Your answer to those prompts based on your query results.
3. Your suggestion to the dealership based on your query results.

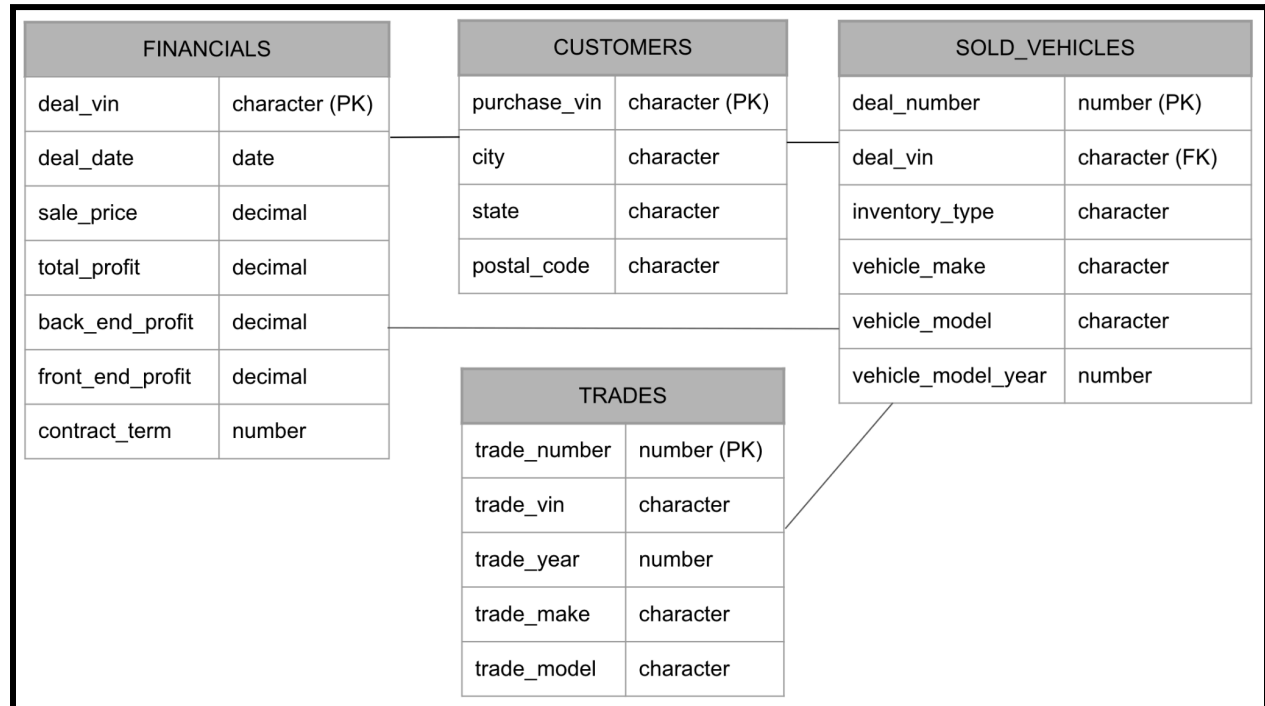
Resources For Final

1. Luxury_Car_Dlrshp.db
2. Luxury Dealership Business Use Case.pdf
3. EoC2 Final Notebook

Data Source:

The database is named 'Luxury_Car_Dlrshp' and contains in total 8,208 rows. There are four tables in the database: 'customers', 'sold_vehicles', 'financials', and 'trades'.

Diagram 1: Luxury_Car_Dlrshp ERD



Constraints:

Regarding 'sold_vehicles' and 'trades' tables:

The 'deal_vin' column from the 'sold_vehicles' table and 'trade_vin' column from the 'trades' table do not share a primary key or foreign key relationship.

To create a relationship between these two tables use 'deal_number' from the 'sold_vehicles' table and 'trade_number' from the 'trades' table since they share primary keys.

Definitions:

sale_price is the cost of the vehicle a customer pays.

total_profit is the total cost the dealership made off of the sale of a vehicle. This is produced by taking the sale price +/- back_end_profit and front_end_profit = total_profit.

back_end_profit is the +/- sales transaction from the service and or parts department.

front_end_profit is the +/- sales transaction from the warranty department and vehicle trade-ins.

Revenue is the total amount of money a business generates from its core operations before any expenses are deducted.

Cost - any expense taken on by the dealership at time of sale to complete the deal for a customer. This expense is typically seen in the `back\_end\_profit` observations related to repairs or replacements made on a pre-owned (used) vehicle being prepped for sale.

New (N) - is a vehicle that has never been titled to a consumer. It is sold as an original asset directly from the manufacturer to the dealership.

Used (U) - is a vehicle that has had at least one previous owner and has been previously registered/titled. This vehicle status is also known as “pre-owned” or if the vehicle is authorized as meeting the inspection criteria for an extended warranty, offered by the brand manufacturer, then these vehicles are often referred to as “certified”.